

Alpha capture programs provide managers with answers to many of their investment related questions: How much alpha will my investment achieve? How much should I invest? And most importantly, how is transaction cost analysis being utilized to analyze profitability concerns?

To accurately compute expected alpha capture, managers need to specify their alpha estimates and have accurate market impact modeling capabilities. And this is yet another reason why TCA has historically gained so little traction in the industry. Portfolio managers are unwilling to provide brokers with their alpha estimates, and brokers have been unwilling to provide managers with the underlying market impact models.

Market impact and alpha cost are conflicting terms. Trading too fast will incur too much impact but trading too slow will miss too much alpha (or missed profit opportunity). Subsequently, we refer to this conflicting expression as the portfolio manager's dilemma.

In Chapter 5 we provided techniques to develop and test a market impact model. In this chapter we provided a pre-trade of pre-trade model to decipher broker models, to calibrate preferred market impact parameters, and allow those models to function as a stand-alone application on the investor's own desktop and as part of their own in-house proprietary systems.

The pre-trade of pre-trades approach is the easiest way for analysts and managers to solve the issue of not having a market impact model on their desktop. The process provides managers with a functional form of a model and also allows them to run that model from their own desktop. They can then incorporate their own liquidity and volatility views, and perform sensitivity analysis with different alpha estimates and various different market conditions. This last piece is the most important since it allows managers to keep their alpha expectations proprietary. Imagine if fund managers did provide their brokers with their alpha views!

Example 5

Portfolio managers develop alpha capture programs by incorporating the simplified I-Star impact model and the manager's alpha forecast. For a continuous trading strategy and an alpha estimate following a linear trend, the manager will incur an alpha cost equal to one-half the total alpha movement over the trading period t . This mathematical representation is:

$$\text{Alpha Cost}_{bp} = \frac{1}{2} \cdot \frac{\mu_{bp}}{d} \cdot t \quad (11.19)$$

Where μ_{bp} is the alpha forecast, d is the time horizon of the alpha forecast, and t is the time to complete the order with the condition $0 \leq t \leq 2$.